

General Terminology

Descriptive statistics	Summarizing, organizing and describing data.
Inferential statistics	using probability theory to extrapolate conclusions from data.
Population	A collection of all individuals or items of interest.
Sample	A subset of the population.
statistic	A numerical property of a sample.
Parameter	A numerical property of the whole population.
Variable	A characteristic or measurement of interest for the individuals in a population.
Data	The values of the variables.
Quantitative variable	A variable that can be measured numerically.
Qualitative variable	A variable that can be categorized based on some characteristic or attribute.
Discrete variable	A quantitative variable that can take on only a finite or countable number of values.
Continuous variable	A quantitative variable that can take on any value in an interval of real numbers.
Nominal variable	A qualitative variable that has no natural ordering.
Ordinal variable	A qualitative variable that has a natural ordering.
Frequency	The number of times a value or category occurs.
Relative frequency	Normalized frequency, expressed as a percentage.
frequency table	A table that displays the frequency or relative frequency of each value.
Rounding	Carry the final answer with one more decimal place than the data with the least number of decimal places.

Sampling Methods

There are 2 main types of sampling methods: random and non-random.

Simple random	Every individual in the population has an equal chance o being selected.
Stratified	The population is divided into subgroups (strata) based on some characteristic, and a random sample is taken from each stratum.
Cluster	The population is divided into clusters, and a random sample of clusters is selected. All individuals in the selected clusters are included in the sample.
Systematic	A random starting point is selected, and then every k^{th} individual is selected from the population.
Convenience	The sample is selected based on convenience or accessibility.

Levels of Measurement

Nominal	Nornumerically meaningful(labels).
Ordinal	Can order (star rating).
Interval	Distance between values is meaningful.
Ratio	Has a definite zero, ratios are meaningful (e.g., height, time)

General Tips

- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ • $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ • $\sum_{i=1}^n r^i = \frac{r(1-r^n)}{1-r}$ for $r \neq 1$ • $\sqrt{x^2} = |x|, x \in \mathbb{R}$ • $(\sqrt{x})^2 = x, x \geq 0$ • $\cos \theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$
- $\sin \theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$ • $e^{i\theta} = \cos \theta + i \sin \theta$ • $(a+b)^3 = a^3+b^3+3ab(a+b)$ • $(a-b)^3 = a^3-b^3-3ab(a-b)$ • $a^3+b^3 = (a+b)(a^2-ab+b^2)$
- $a^3-b^3 = (a-b)(a^2+ab+b^2)$

Stat Formulas

Interquartile range(IQR)	$IQR = Q_3 - Q_1$
Range	$R = x_{max} - x_{min}$
Percentile Position	$P_k = \frac{k(n+1)}{100}$
Sample Mean	$\bar{x} \approx \frac{\sum_{i=1}^n x_i}{n}$
Population Mean	μ
Median	The middle value of a sorted data set.
Mode	The most frequently occurring value in a data set.
Range	$R = x_{max} - x_{min}$
Sample Variance	$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
Population Variance	$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$
Standard deviation	$\sigma = \sqrt{\sigma^2}$
Z-score	$z = \frac{x - \bar{x}}{\sigma}$
Skewness	$Sk = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$

Probabilities

A probability quantifies the chance that an event occurs. A probability must respect the kolmogorov axioms listed below.

- $\mathbb{P}(A) \geq 0, \forall A \in S$
 - $\mathbb{P}(S) = 1$
 - $\mathbb{P}(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mathbb{P}(A_i)$ if $A_i \cap A_j = \emptyset$ for all $i \neq j$.
- | | |
|-------------------------|--|
| Experiment | A process that leads to one of several possible outcomes. |
| Outcome | The result of a single trial of an experiment. |
| Event | A collection of one or more outcomes. |
| Sample space | The set of all possible outcomes of an experiment. Notatied S . |
| Equally likely outcomes | $\mathbb{P}(A) = \frac{ A }{ S }$ |
| Expected value | $\mathbb{E}(X) \equiv \sum_{i=1}^n x_i \mathbb{P}(x_i) = \mu$ |
| Conditional probability | $\mathbb{P}(A B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \mathbb{P}(A)$ given B occured. |
| Additive rule | $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ |
| Multiplicative rule | $\mathbb{P}(A \cap B) = \mathbb{P}(A B)\mathbb{P}(B)$ |
| Complement inverse | $\mathbb{P}(A) + \mathbb{P}(A') = 1$ |

Independent events

knowing one occurred does not change the chance of the other.

- $\mathbb{P}(A|B) = \mathbb{P}(A)$
- $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$

Mutually exclusive events

events cannot occur together.

- $A \cap B = \emptyset$
- $\mathbb{P}(A \cap B) = 0$

Sample space representations

listing, tree diagram, Venn diagram.

Contingency tables